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Firm Volatility in Granular Networks

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1 Big picture

- **Research question.** Why do firm-level volatilities differ so much across firms and move together so strongly over time? More specifically, can the firm size distribution and customer-supplier networks jointly explain the time-series and cross-sectional behavior of firm volatility?
- **Main idea.** Firms are not isolated units. A supplier is exposed not only to its own shocks but also to shocks hitting its customers and its customers' customers. If the customer base is concentrated, shocks are poorly diversified and supplier volatility is high.
- **Core economic mechanism.** Economy-wide size dispersion matters because firms draw customers from the same highly skewed size distribution. When the firm size distribution becomes more unequal, a typical customer network becomes less diversified, raising average firm volatility and the dispersion of firm volatility.
- **Key empirical facts.** Market-based firm size dispersion is strongly correlated with both average firm variance and the cross-sectional dispersion of firm variance. The same fact appears using sales-based measures of size and volatility.
- **Cross-sectional prediction.** Smaller firms should be more volatile because they have fewer customers and therefore less external diversification. Firms with more concentrated customer bases should also be more volatile.
- **Model.** A parsimonious network model lets shocks travel upstream from customers to suppliers. Large suppliers have more customers; large customers receive larger link weights in suppliers' customer portfolios.
- **Quantitative result.** The simulated method of moments estimation finds strong network propagation, with the benchmark network parameter $\gamma = 0.918$. Without network effects, the model cannot generate the observed cross-sectional dispersion in firm volatility.
- **Additional evidence.** Panel regressions show that firm variance is negatively related to firm size and positively related to customer-network concentration. Factor regressions show that firm size dispersion is an important common factor in firm-level variances.

Economic takeaway. Firm-level volatility is partly an endogenous network outcome. Granularity and sparsity do not only matter for aggregate volatility; they also shape each firm's own volatility through the diversification, or lack of diversification, of its customer network.

2 Incremental contribution of the paper

- **Moves the granularity literature to the firm-volatility level.** Prior work emphasizes how skewed firm or sector sizes make idiosyncratic shocks aggregate. This paper asks how the same economic forces determine the volatility of individual firms.
- **Links two distributions.** The paper documents a new time-series relationship between the firm size distribution and the firm volatility distribution: size dispersion predicts both mean volatility and volatility dispersion.
- **Connects firm networks to firm volatility.** Rather than treating heteroscedastic firm shocks as primitive, the paper derives heterogeneity in firm volatility from the customer-supplier network.

- **Uses both market and fundamental volatility.** The main facts hold using stock-return variance and sales-growth variance. This makes it harder to explain the results only through financial-market leverage or discount-rate channels.
- **Combines theory, moments, and micro regressions.** The paper first derives analytic predictions, then estimates a structural network model, and finally tests additional micro implications in the data.
- **Quantifies the network mechanism.** The SMM exercise targets size, volatility, and network moments jointly. The model is not only qualitatively consistent with the facts; it is asked to fit the joint distribution of sizes, variances, and customer-supplier links.
- **Clarifies upstream transmission.** The evidence favors shocks traveling from customers to suppliers: out-Herfindahl matters for volatility, while in-Herfindahl does not have the same explanatory power.
- **Explains volatility comovement.** Firm size dispersion is proposed as an economic factor behind the common factor in firm-level volatilities.

3 Data and measurement

3.1 Firm size and firm volatility

The paper uses both market-based and fundamentals-based measures of size and volatility.

- **Market size.** The main size measure is equity market value at the end of the calendar year.
- **Market variance.** The main variance measure is the variance of daily stock returns within the calendar year.
- **Fundamental size.** The fundamentals-based size measure is annual firm sales.
- **Fundamental variance.** The fundamentals-based variance measure is the variance of quarterly sales growth, using 20 quarters of same-quarter sales growth observations.
- **Sample.** Stock-return data come from CRSP for 1926–2016. Sales data come from the CRSP/Compustat merged file for 1952–2016, with fundamentals-based variance available later because it needs rolling quarterly growth observations.
- **Distributional summary.** The cross-sectional size and variance distributions are summarized by the mean and standard deviation of log quantities. The word *dispersion* refers to the cross-sectional standard deviation.

Interpretation. The empirical object is not only whether firms are volatile on average. It is the joint evolution of the entire distribution of size and the entire distribution of firm variance.

3.2 Customer-supplier network data

The network data come from Compustat segment data for 1980–2012.

- A firm is required to report customers that account for more than 10% of its sales.

- The data identify a set of supplier-customer links among public firms and provide sales shares for many links.
- In a typical year, the paper has about 1,330 suppliers with nonmissing customer information.
- Because public-firm data are incomplete and small customer links are truncated, the estimation explicitly models both selection and truncation.

3.3 Network variables

For supplier i and customer j , w_{ij} is the weight of customer j in supplier i 's customer base.

- **Out-degree** N_i^{out} . Number of customers of supplier i .
- **Out-Herfindahl** H_i^{out} . Customer-base concentration, $H_i^{out} = \sum_j w_{ij}^2$. A high value means that a supplier depends heavily on a small number of customers.
- **In-degree** N_i^{in} . Number of suppliers serving firm i , inferred from the customer data.
- **In-Herfindahl** H_i^{in} . Concentration of firm i 's supplier network.

Interpretation. The out-Herfindahl is the key network statistic for upstream shock propagation. If customer shocks matter for suppliers, then concentration among a supplier's customers should raise the supplier's volatility.

4 Methodology and empirical specifications

4.1 Main time-series facts

For each year, the paper computes cross-sectional moments of log firm size and log firm variance. The central facts are correlations between lagged size dispersion and volatility moments.

- In the market-based data, lagged size dispersion has a 63.3% correlation with average log firm variance.
- In the market-based data, lagged size dispersion has a 77.6% correlation with the dispersion of log firm variance.
- In the sales-based data, lagged size dispersion has a 68.7% correlation with average log sales-growth variance.
- In the sales-based data, lagged size dispersion has an 81.7% correlation with sales-growth variance dispersion.

Interpretation. When firm sizes become more dispersed, firm volatility rises broadly and volatility becomes more unequal across firms.

4.2 Reduced-form network growth model

Firm i 's growth rate is affected by its own idiosyncratic shock and by the growth rates of its customers:

$$g_i = \mu_g + \gamma_i \sum_{j=1}^N w_{ij} g_j + \varepsilon_i. \quad (1)$$

In vector notation, setting $\mu_g = 0$,

$$g = \Gamma W g + \varepsilon = (I - \Gamma W)^{-1} \varepsilon. \quad (2)$$

The matrix $(I - \Gamma W)^{-1}$ is the Leontief inverse and describes how shocks propagate through the network.

Interpretation. The model is a spatial autoregression on the customer-supplier network. A customer shock affects its supplier directly, and then affects other firms indirectly through higher-order network paths.

4.3 Structural interpretation

The appendix provides a production-economy interpretation. There are multiple firms producing goods, a representative household with Cobb-Douglas preferences, and firms that use other firms' outputs as inputs. Taste shocks to final demand travel upstream from customers to suppliers. Under the model, changes in firm output satisfy

$$\frac{dY}{Y} = (I - \Gamma W)^{-1} \frac{d\eta}{\omega}. \quad (3)$$

Interpretation. The reduced-form network equation can be viewed as the equilibrium response of firm output or sales to demand shocks in an interconnected production economy.

4.4 Size effects in network formation

The model uses two size-based assumptions.

- **Large suppliers have more customers.** The probability that supplier i connects to any customer depends positively on supplier size:

$$P(b_{ij} = 1) = p_i = \frac{\tilde{S}_i}{Z} N^{-\zeta}. \quad (4)$$

This implies expected out-degree

$$N_i^{out} \approx N p_i = \frac{\tilde{S}_i}{Z} N^{1-\zeta}. \quad (5)$$

- **Large customers receive larger weights.** Conditional on a link, customer size determines link strength:

$$w_{ij} = \frac{b_{ij} S_j}{\sum_k b_{ik} S_k}. \quad (6)$$

In the estimated model, this becomes

$$w_{ij} = \frac{b_{ij}S_j^\psi}{\sum_k b_{ik}S_k^\psi}. \quad (7)$$

Interpretation. Large suppliers have broader customer portfolios. Large customers are more important to their suppliers. These two assumptions map the firm size distribution into the network structure.

4.5 Variance and diversification logic

A first-order approximation gives a simple expression for firm variance:

$$V(g_i) \approx \sigma_\varepsilon^2 (1 + \gamma_i^2 H_i^{out}). \quad (8)$$

- More customers lower H_i^{out} , improving diversification.
- More equal customer sizes lower H_i^{out} , also improving diversification.
- Higher economy-wide size dispersion makes every firm's customer draw more unequal in expectation, raising firm volatilities together.

The full proposition shows that firm variance is increasing in the economy-wide size-dispersion term $E[S^2]/E[S]^2$, and that smaller firms have larger exposure to this common volatility factor because they have fewer customer links.

4.6 SMM estimation

The estimated model adds internal diversification: larger firms have lower fundamental shock volatility,

$$\sigma_{\varepsilon,i} = \sigma_\varepsilon + \lambda \log \left(1 + \frac{S_{median}}{S_i} \right). \quad (9)$$

The simulated method of moments chooses parameters to match 28 moments of the distributions of firm size, firm variance, and network linkages.

- Estimated parameters include γ , σ_ε , λ , Z , ψ , μ_s , σ_s , and N_{pub} .
- The sparsity parameter is set to $\zeta = 0.87$ based on the elasticity of connections with respect to the number of firms.
- The model is simulated for 5,000 firms, but moments are computed for the largest N_{pub} firms to mimic the public-firm sample.
- Customer links below 10% of supplier sales are treated as unobserved to match the Compustat reporting threshold.

4.7 Additional testable predictions

The paper then tests two model implications directly.

- **Determinants of firm volatility.** Firm variance should fall with firm size and rise with customer-base concentration.
- **Comovement in firm volatilities.** Firm size dispersion should act as a common factor in firm-level volatility, with larger firms having lower exposure to this factor.

5 Identification and interpretation

5.1 What the paper identifies

The paper identifies a tight empirical relationship between size dispersion, network concentration, and firm volatility. It also shows that a network model with upstream propagation can quantitatively match several key moments of the data.

The evidence is strongest as a combination of reduced-form facts and structural discipline. The paper does not randomly assign network links or firm sizes, so the interpretation relies on the model's network formation assumptions and on robustness checks designed to rule out simpler explanations.

5.2 Why the market and fundamental evidence matters

The facts appear in both stock-return volatility and sales-growth volatility. This is important because purely financial explanations, such as leverage or discount-rate variation, have difficulty explaining why fundamentals-based volatility moments move with the size distribution in the same way.

5.3 Selection and truncation in network data

The Compustat segment data only observe public firms with nonmissing customer information and generally only customers representing at least 10% of supplier sales. The estimation explicitly models both issues:

- selection into the public/network-data sample is approximated by computing moments for the largest public firms in the simulated economy;
- truncation is approximated by deleting simulated customer links with weights below 10%.

5.4 Interpreting the large network parameter

The benchmark estimate $\gamma = 0.918$ implies very strong propagation. Interpreted literally in the structural production model, this would suggest a very small final-goods share, which is implausibly low relative to input-output data. The paper argues that this may reflect broader links beyond physical input-output relationships or heterogeneity in γ_i that is correlated with centrality. If

central firms have high propagation parameters, a homogeneous- γ estimation can recover a high common γ .

5.5 Composition concerns

A possible alternative explanation is that the composition of public firms changes over time, especially as small volatile firms enter or leave public markets. The paper addresses this in three ways:

- Table 1 shows the main correlations across size groups, industries, ages, exchanges, and subperiods.
- The facts are visible both before and after 1970.
- Appendix exercises show that a changing-public-firm-composition model without network effects has difficulty generating the observed movements in mean volatility and volatility dispersion.

5.6 Upstream versus downstream propagation

The out-Herfindahl of a firm's customer network predicts its volatility. In contrast, the in-Herfindahl of its supplier network is not similarly important. This pattern supports upstream propagation from customers to suppliers rather than downstream propagation from suppliers to customers.

6 Tables and figures

6.1 Figures 1 and 2: Size dispersion and variance moments

Read:

- Figure 1 uses market equity for firm size and daily stock-return variance for firm variance.
- Panel A shows that lagged market size dispersion moves closely with average log variance.
- Panel B shows that lagged market size dispersion also moves closely with the dispersion of log variance.
- Figure 2 repeats the exercise using sales for size and sales-growth variance for volatility.
- The fundamentals-based series confirm the same positive association.

Economic message: The central fact is not only that small firms are more volatile. The whole volatility distribution comoves with the whole size distribution. When firm sizes become more unequal, firm volatility becomes higher and more dispersed.

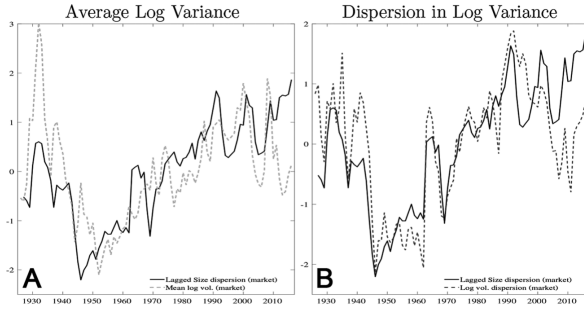


FIG. 1.—Market-based dispersion in firm size and firm variance. Panel A plots the lagged cross-sectional dispersion of log firm size (solid line) and the cross-sectional mean of the log variance distribution (dashed line). Panel B plots the lagged cross-sectional dispersion of log firm size (solid line) and the cross-sectional standard deviation of the log variance distribution (dashed line). Firm size is measured as market value of equity; firm variance is measured as the variance of daily stock returns within the year. All series are rescaled for the figure to have mean 0 and variance 1. Sample is from 1926 to 2016 at annual frequency. A color version of this figure is available online.

Figure 1: Market-based dispersion in firm size and firm variance.

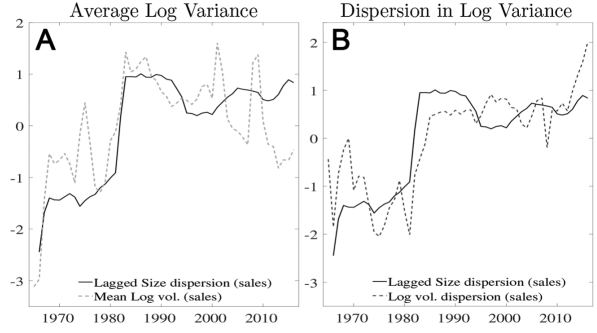


FIG. 2.—Fundamentals-based dispersion in firm size and firm variance. Panel A plots the lagged cross-sectional dispersion of log firm size (solid line) and the cross-sectional mean of the log variance distribution (dashed line). Panel B plots the lagged cross-sectional dispersion of log firm size (solid line) and the cross-sectional standard deviation of the log variance distribution (dashed line). Firm size is measured as firm sales; firm variance is measured based on 20 quarters of growth in firm sales. All series are rescaled for the figure to have mean 0 and variance 1. Sample is from 1965 to 2016 at annual frequency. A color version of this figure is available online.

Figure 2: Fundamentals-based dispersion in firm size and firm variance.

6.2 Tables 1 and 2: Composition facts and estimated parameters

Read Table 1:

- For all firms, the correlation between size dispersion and average variance is 67.5%.
- For all firms, the correlation between size dispersion and variance dispersion is 77.4%.
- The positive correlations hold within size terciles, most industries, firms of different public ages, both main subperiods, and both NYSE and non-NYSE samples.
- Utilities are the main weak case for average variance, but the variance-dispersion correlation remains positive.

Read Table 2:

- The benchmark network propagation estimate is $\gamma = 0.918$, with a small standard error.
- The fundamental shock volatility parameter is 0.298, and internal diversification is governed by $\lambda = 0.131$.
- The connection-weight curvature estimate is $\psi = 0.184$, meaning link weights increase with customer size but less than one-for-one.
- The model selects about 1,332 public firms, matching the network-data sample size.

Economic message: Table 1 establishes that the empirical fact is broad rather than driven by one subsample. Table 2 shows that fitting the volatility distribution requires strong network propagation plus internal diversification.

TABLE 1
COMPOSITION

	NUMBER OF FIRMS (1)	CORRELATION OF SIZE DISPERSION WITH	
		Average Variance (2)	Dispersion Variance (3)
All firms	3,069.9	67.5	77.4
By size:			
Smallest third	1,023.4	65.8	59.9
Middle third	1,023.0	59.3	67.2
Largest third	1,023.5	55.0	66.4
By industry:			
Consumer nondurables	240.8	59.5	67.2
Consumer durables	104.6	34.9	71.7
Manufacturing	516.4	51.7	77.3
Energy	150.2	67.7	57.4
Technology/ telecommunications	494.4	73.9	51.0
Retail	305.8	62.1	63.4
Health care	199.0	73.6	53.6
Utilities	111.5	16.9	58.9
Other	884.0	56.6	66.2
By time since entry:			
Less than 5 years	905.0	74.0	79.3
At least 5 years	2,159.3	71.5	77.1
At least 10 years	1,447.5	72.9	76.4
At least 25 years	489.0	46.9	77.9
By subsample:			
1926–69	496.7	74.5	83.8
1970–2016	2,573.1	55.2	42.7
By exchange:			
NYSE	1,204.3	60.8	76.6
Non-NYSE	1,865.6	41.3	42.4

NOTE.—Annual data are from 1926–2016. We use the market-based volatility measure constructed from stock returns and the market-based measure of size (market equity). Column 1 reports the time series average of the cross section number of firms. Column 2 reports the correlation between average log volatility ($\mu_{e,t}$) and lagged log size dispersion ($\sigma_{s,t-1}$). Column 3 reports the correlation between dispersion in log volatility ($\sigma_{e,t}$) and lagged log size dispersion. For each grouping, we first compute the cross-sectional moments among the firms in the group, and then we compute the time series correlation between the moments.

Table 1: Correlations by composition group.

TABLE 2
PARAMETER ESTIMATES

Parameter	Definition	Benchmark (1)	No Network (2)	No Internal Diversification (3)
γ	Network propagation effect	.918 (.026)	.000 ...	.918 ...
σ_e	Fundamental shock volatility	.298 (.035)	.298 ...	.388 ...
λ	Internal diversification effect	.131 (.025)	.131 ...	.000 ...
Z	New connections	.003 (.001)	.003 ...	.003 ...
ψ	Sensitivity of connection weight to customer size	.184 (.003)	.184 ...	.184 ...
μ_s	Mean of initial log size distribution	13.412 (.243)	13.412 ...	13.412 ...
σ_s	Standard deviation of initial log size distribution	5.104 (.102)	5.104 ...	5.104 ...
N_{pub}	Number of public firms	1,332 (47.7)	1,332 ...	1,332 ...

NOTE.—This table report model parameters. Column 1 reports the estimated parameters using SMM estimation along with standard errors in parentheses. Columns 2 and 3 report the parameters from the benchmark estimation but without network effects (i.e., $\sigma = 0$) and without internal diversification (i.e., $\lambda = 0$), respectively.

Table 2: Benchmark parameter estimates.

6.3 Tables 3 and 4: Model fit and firm-level volatility regressions

Read Table 3:

- The benchmark model matches the mean of log variance and generates a standard deviation of log variance of 0.86, compared with 1.05 in the data.
- The no-network model generates almost no dispersion in log variance: standard deviation 0.01.
- The no-internal-diversification model generates only about half the observed dispersion: standard deviation 0.46.
- The benchmark model matches the signs and rough magnitudes of key network correlations and has a Wald-test p -value of 0.64.

Read Table 4:

- Log size enters negatively and strongly across specifications.
- Log out-Herfindahl enters positively, even with controls for age, leverage, industry concentration, cohort fixed effects, and industry fixed effects.
- The size coefficient is around -0.15 , while the out-Herfindahl coefficient is around 0.02 – 0.06 .
- This supports the model’s prediction that smaller firms and firms with more concentrated customer bases have higher volatility.

Economic message: Table 3 is the quantitative model discipline: without networks the model

fails. Table 4 is the micro evidence: the same network variables that matter in the model also predict firm volatility in the data.

TABLE 3
SIZE, VARIANCE, AND NETWORK MOMENTS

	MODEL			
	DATA (1)	Bench (2)	No Network (3)	No Internal Diversification (4)
A. Size and Volatility Distributions				
log size:				
Average	19.83	19.83	19.83	19.83
Standard deviation	2.56	2.53	2.53	2.53
50th–10th percentile	2.91	2.37	2.37	2.37
90th–50th percentile	3.88	3.91	3.91	3.91
log variance:				
Average	−1.40	−1.40	−2.42	−1.39
Standard deviation	1.05	.86	.01	.46
50th–10th percentile	1.38	.40	.00	.24
90th–50th percentile	1.31	1.69	.02	.93
B. Network Moments				
N^{out} :				
Median	1.00	1.12	1.12	1.12
log 50th–10th percentile	.00	.08	.08	.08
log 90th–50th percentile	.83	1.02	1.02	1.02
H^{out} :				
Median	.05	.18	.18	.18
log 50th–10th percentile	1.27	1.88	1.88	1.88
log 90th–50th percentile	1.85	1.71	1.71	1.71
N^{in} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.00	.00	.00	.00
log 90th–50th percentile	1.69	1.09	1.09	1.09
H^{in} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.95	.89	.89	.89
log 90th–50th percentile	.00	.00	.00	.00
C. Cross-Sectional Correlations				
Corr(log size, log var)	−.64	−.61	−.67	−.69
Corr(log N^{out} , log size)	−.09	.31	.31	.31
Corr(log H^{out} , log size)	−.28	−.88	−.88	−.88
Corr(log N^{in} , log size)	.22	.51	.68	.75
Corr(log H^{in} , log size)	.53	.34	.34	.34
Corr(log N^{out} , log var)	−.47	−.34	−.34	−.34
Corr(log H^{out} , log var)	.13	.19	.19	.21
D. Firms with Network Data				
log number of public firms	7.15	6.95	6.95	6.95
E. Goodness of Fit Statistics				
Function value	...	2.88	7.53	3.72
Wald	...	24.76	...	...
p-value	...	.64	...	...

NOTE.—This table reports different size, variance, and network moments both from the data and from our simulation. We use Compustat and CRSP data from 1980 to 2012, and the moments reported are time series averages of cross-sectional moments. Column 2 reports the moments from an SMM estimation of the benchmark model. Columns 3 and 4 report the SMM estimation of a model without network effects (i.e., $\gamma = 0$) and without internal diversification effects (i.e., $\lambda = 0$), respectively. Panel A reports average, standard deviation, and percentile spreads for both log size and log variance. Panel B reports median and percentile spreads for four distinct network moments: out-degree or number of customers (N^{out}), out-Herfindahl (H^{out}), in-degree or number of suppliers (N^{in}), and in-Herfindahl (H^{in}). Panel C reports correlations between log size, log variance, and network moments. Panel D reports the log number of firms with network data available. Finally, in panel E, we report the distance function minimized in the SSM and a Wald test for the null hypotheses that all moments in the data (col. 1) and in the estimated model (col. 2) are equal.

Table 3: Size, variance, and network moments in data and model.

TABLE 4
DETERMINANTS OF FIRM-LEVEL VOLATILITY (Dependent Variable: log Firm Volatility)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
log size	−.16		−.15	−.15	−.15	−.15	−.14		−.15	−.15	−.15	−.15
log H^{out}	−16.98	.06	−19.67	−28.68	.05	−18.38	−18.45	.05	−19.02	−20.13	−19.60	−19.60
log age		4.22	2.63		4.46	2.17		4.44	1.96		18.67	.02
Leverage				−.09	−.09	−.09	−.09					
Industry concentration				−2.49	−2.49	−2.49	−2.49					
Constant	−1.00	−.57	−.80	−.92	−.92	−.92	−.92	−.55	−1.15	−.46	−.38	−.41
Fixed effects	−19.70	−8.82	−11.81	−5.88	−.05	−4.56	−10.89	−4.46	−8.48	−26.85	−10.66	−18.94
Observation	None	None	None	None	None	None	Cohort	Cohort	Cohort	Industry	Industry	Industry
Adjusted R^2	167,440	28,718	28,232	133,026	22,070	22,940	186,479	23,084	25,065	167,440	28,718	28,232
	.283	.028	.299	.338	.333	.373	.358	.145	.306	.307	.137	.379

NOTE.—This table reports panel regressions of firms' return volatility on a range of characteristics, including log size, log customer network Herfindahl (H^{out}), log age, book leverage, competition measured by industry size concentration, as well as cohort and industry fixed effects in certain specifications. Size is defined as the lagged market equity relative to the cross-sectional average market equity. Book leverage is defined as debt in current liabilities plus total long-term debt divided by total assets. Industries are defined as four-digit Global Industry Classification Standard codes, and industry concentration is measured by the Herfindahl index of sales. Cohorts are defined by the year in which a firm first appears in the CRSP/Compustat data set. Standard errors are clustered at the industry level. Sample is at annual frequency from 1980 to 2012.

Table 4: Determinants of firm-level volatility.

6.4 Table 5 and Figure A1: Volatility comovement and structural support

Read Table 5:

- Lagged size dispersion explains 15.65% of firm-level return-volatility variation on average in the data.
- Lagged mean volatility explains 20.94%, and contemporaneous mean volatility explains 33.65%.
- Small firms have larger loadings on the size-dispersion factor than large firms: 0.71 for Q1 versus 0.37 for Q5.
- The model reproduces the prediction that size dispersion is a strong volatility factor and that large firms have smaller exposure.

Read Figure A1:

- Figure A1 comes from the structural appendix.
- It plots average firm variance against the size Herfindahl in simulated three-firm economies.
- For the vast majority of simulated models, higher size concentration increases average firm variance.

Economic message: The paper’s common-factor claim is disciplined in two ways. Table 5 shows it in the data and model; Figure A1 shows that the underlying size-dispersion-to-volatility logic also appears in the structural model.

TABLE 5 COMOVEMENT IN FIRM VARIANCES						
	FACTORS					
	Data			Model		
	$\sigma_{S,t-1}$ (1)	$\mu_{S,t-1}$ (2)	$\mu_{S,t}$ (3)	$\sigma_{S,t-1}$ (4)	$\mu_{S,t-1}$ (5)	$\mu_{S,t}$ (6)
A. Return Volatility R^2 by Size Quintile						
All firms	15.65	20.94	33.65	52.18	55.92	93.22
Q1	19.44	22.85	29.37	51.99	54.97	84.14
Q5	11.06	17.62	36.55	51.91	56.10	98.48
B. Return Volatility Loadings by Size Quintile						
All firms	.55	.48	.73	1.31	.67	1.06
Q1	.71	.57	.73	1.12	.57	1.15
Q5	.37	.43	.76	.31	.16	.24

NOTE.—Each factor model regression is a time series regression of log total volatility on a factor. Total volatility is measured as variance of daily returns within the calendar year, and size is measured as market equity. All volatility factor regressions take the form $\log \sigma_{i,t} = a_i + b_i \text{factor}_t + e_{i,t}$. For each stock i , we estimate the factor model and report the cross-sectional average of the R^2 in panel A and the cross-sectional average of the slopes in panel B. We require a minimum of 25 observations to run the regression. In each panel, we report the full sample average and the average within the first and last quintiles of the time series average size distribution. We deflate size by consumer price index when constructing size quintiles. We consider three different volatility factors. The first, motivated by our network model, is the lagged cross-sectional standard deviation of log market equity, $\sigma_{S,t-1}$ (cols. 1, 4). The second and third factors we consider are the lagged and contemporaneous cross-sectional average log volatility, $\mu_{S,t-1}$ (cols. 2, 5) and $\mu_{S,t}$ (cols. 3, 6). All three factors use the full sample in the cross section. Columns 1–3 report the estimation results for the data at annual frequency from 1926 to 2016. We estimate the same factor structure for the model as well in cols. 4–6.

firm size dispersion is about as strong a predictor as lagged average firm variance. The R^2 is about half that of contemporaneous mean variance.¹⁸

Table 5: Comovement in firm variances.

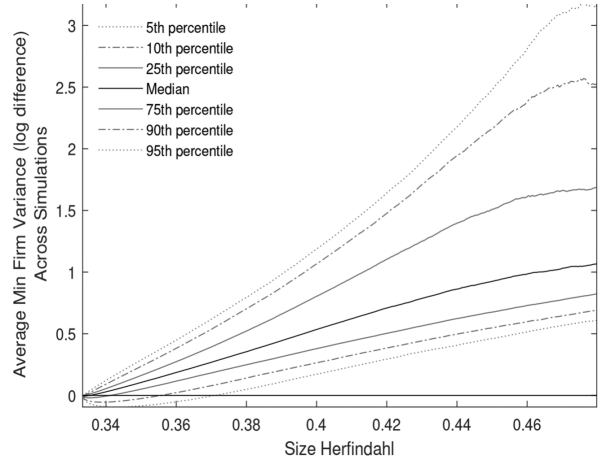


FIG. A1.—Minimum average firm variance by size Herfindahl for 50,000 network models. The figure plots the average firm variance as size Herfindahl increases for 50,000 simulated networks. For each simulation, we find the feasible firm size distribution that delivers the size Herfindahl displayed in the X-axis and the minimum average firm variance. Specifically, in each simulation, we assume that there are three firms, and we randomly draw the entries of the matrix A from a uniform distribution in the unit interval. We require each simulated network to satisfy two properties: (1) all rows in A sum to less than or equal to 1 and (2) there has to be a distribution of preference parameters such that firms having equal shares are an equilibrium. Once we draw a feasible network, we then solve for different preference parameters that minimize the average firm variance and deliver the Herfindahl target. Finally, we compute the log difference between the simulated economy relative to the same economy when firms have equal size (i.e., minimum size Herfindahl). A color version of this figure is available online.

Figure A1: Average firm variance by size Herfindahl.

6.5 Figure C1 and Table C1: Business-cycle robustness and truncation

Read Figure C1:

- The figure applies an HP filter and plots detrended moments of size and volatility distributions.
- The same positive relationship appears at business-cycle frequencies, not only in long-run trends.

Read Table C1:

- Table C1 studies truncation in customer-supplier network data.
- The observed Compustat network is truncated by reporting rules, which biases some correlations downward.

- Untruncated BEA data and the model both support the assumptions that large suppliers have more customers and that large customers receive larger weights.

Economic message: Appendix C strengthens two pieces of the argument: the size-volatility fact is not just a slow trend, and the estimated network assumptions are plausible after accounting for truncation.

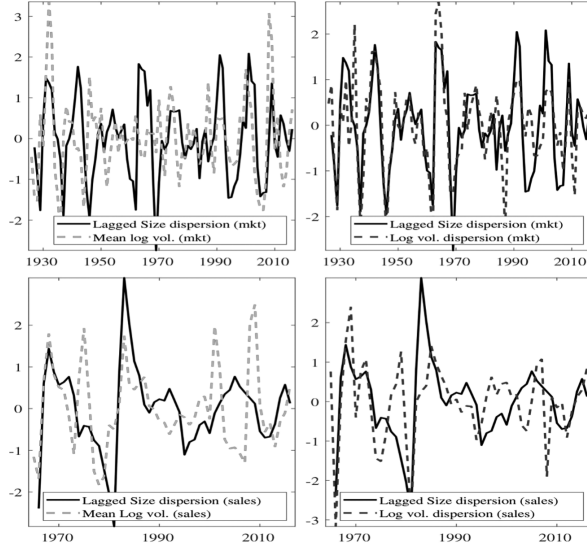


FIG. C1.—Detrended size and variance moments. The figure plots HP-detrended time series moments of size and volatility distributions using a smoothing parameter of 50. A
Figure C1: Detrended size and variance moments.

6.6 Tables C2 and C3: Re-estimating restricted models

Read Tables C2 and C3:

- Table C2 re-estimates the no-network and no-internal-diversification models rather than simply imposing restrictions on benchmark parameters.
- When $\gamma = 0$, the estimation raises fundamental volatility, but it still cannot generate meaningful cross-sectional volatility dispersion.
- When $\lambda = 0$, the estimation raises γ , but the model still falls short on volatility dispersion.
- Table C3 shows the corresponding moments, confirming that both network effects and internal diversification are needed.

Economic message: The conclusion that networks are necessary is not an artifact of holding other parameters fixed. Even after re-estimation, restricted models have inferior fit.

TABLE C1
TRUNCATION ANALYSIS

	TRUNCATED			UNTRUNCATED	
	Compustat (1)	BEA (2)	Model (3)	BEA (4)	Model (5)
$\text{Corr}(w_{i,p}, \log S_i)$	.19	.55	1.00	.40	1.00
$\text{Corr}(N_i^{\text{out}}, \log S_i)$	-.00	-.07	.31	.58	.69
$\text{Corr}(H_i^{\text{out}}, \log S_i)$	-.28	-.39	-.88	-.38	-.67
$\text{Corr}(N_i^{\text{in}}, \log S_i)$	.53	.34	.34	.17	-.01
$\text{Corr}(H_i^{\text{in}}, \log S_i)$	-.47	.10	-.34	.10	.14

NOTE.—The table reports time series averages of cross-sectional correlations between various features of customer-supplier networks with size and volatility. We report the following correlations: correlation between the log network weight and log customer size, correlation between the out-degree (number of customers) and log supplier size, correlation between the out-Herfindahl and log supplier size, correlation between the in-degree (number of suppliers) and log supplier size, correlation between the in-Herfindahl and log supplier size, and correlation between the network weight and log size. Column 1 reports the correlation using the Compustat sample for firms with network data available. Compustat sample is at the firm level and annual frequency for the period 1980–2012. Columns 2 and 4 are based on annual industry-level BEA data for a set of 65 consistently measured industries for the period 1997–2015. Column 2 reports the correlation using BEA data assuming and artificially imposing 10% truncation on the network weights, which implies that we discard all customer-supplier pairs that represent less than 10% of supplier sales. Column 4 is not truncated. Columns 3 and 5 report the correlations for the truncated and untruncated model, respectively.

Table C1: Truncation analysis.

TABLE C2
PARAMETER ESTIMATES

Parameter	Definition	Benchmark (1)	No Network (2)	No Internal Diversification (3)
γ	Network propagation effect	.918 (.026)	.000 ...	.949 (.005)
σ_e	Fundamental shock volatility	.298 (.035)	.496 (.025)	.335 (.010)
λ	Internal diversification effect	.131 (.025)	.090 (3.674)	.000 ...

Table C2, part 1: Restricted-model parameter estimates.

TABLE C3
SIZE, VARIANCE, AND NETWORK MOMENTS

	MODEL			
	DATA (1)	Bench (2)	No Network (3)	No Internal Diversification (4)
A. Size and Volatility Distributions				
log size:				
Average	19.83	19.83	19.86	19.83
Standard deviation	2.56	2.53	2.64	2.60
50th–10th percentile	2.91	2.37	2.49	2.56
90th–50th percentile	3.88	3.91	4.05	3.89
log variance:				
Average	−1.40	−1.40	−1.40	−1.40
Standard deviation	1.05	.86	.00	.55
50th–10th percentile	1.38	.40	.00	.34
90th–50th percentile	1.31	1.69	.01	1.07
B. Network Moments				
N^{net} :				
Median	1.00	1.12	1.12	1.00
log 50th–10th percentile	.00	.08	.08	.00
log 90th–50th percentile	.83	1.02	1.02	1.10
H^{net} :				
Median	.05	.18	.19	.24
log 50th–10th percentile	1.27	1.88	1.91	2.00
log 90th–50th percentile	1.85	1.71	1.66	1.44
N^{int} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.00	.00	.00	.00
log 90th–50th percentile	1.69	1.09	1.10	1.09

Table C3, part 1: Restricted-model moments.

TABLE C2 (Continued)

Parameter	Definition	Benchmark (1)	No Network (2)	No Internal Diversification (3)
Z	New connections	.003 (.001)	.002 (.000)	.004 (.000)
ψ	Sensitivity of connection weight to customer size	.184 (.003)	.189 (.005)	.208 (.002)
μ_s	Mean of initial log size distribution	13.412 (.243)	13.192 (.157)	13.384 (.166)
σ_s	Standard deviation of initial log size distribution	5.104 (.102)	5.291 (.035)	5.152 (.052)
N_{pub}	Number of public firms	1,332 (47.7)	1,337 (40.8)	1,388 (46.3)

NOTE.—This table reports model parameters. Column 1 reports the estimated parameters from the SMM. Columns 2 and 3 report the estimated parameters assuming no network effects (i.e., $\gamma = 0$) and no internal diversification (i.e., $\lambda = 0$), respectively. Compared with the exercise in the main text, all model parameters are reestimated in cols. 2 and 3. Standard errors are in parentheses.

Table C2, part 2: Continued restricted-model parameter estimates.

TABLE C3 (Continued)

	MODEL			
	DATA (1)	Bench (2)	No Network (3)	No Internal Diversification (4)
H^{net} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.95	.89	.90	.84
log 90th–50th percentile	.00	.00	.00	.00
C. Cross-Sectional Correlations				
Corr(log size, log var)	−.64	−.61	−.66	−.70
Corr(log N^{net} , log size)	−.00	.31	.31	.35
Corr(log H^{net} , log size)	−.28	−.88	−.89	−.89
Corr(log H^{net} , log var)	.22	.51	.66	.69
Corr(log N^{int} , log size)	.53	.34	.36	.36
Corr(log H^{int} , log size)	−.47	−.34	−.37	−.36
Corr(log H^{int} , log var)	.13	.19	.20	.22
D. Firms with Network Data				
log number of public firms	7.15	6.95	6.95	6.98
E. Goodness of Fit Statistics				
Function value	...	2.88	6.50	3.45
Wald	...	24.76	26.80	27.33
p	...	.64	.53	.50

NOTE.—This table reports different size, variance, and network moments both from the data and from our simulation. It takes the same structure as table 3. The only difference is that all model parameters are reestimated in cols. 3 and 4.

Table C3, part 2: Continued restricted-model moments.

6.7 Tables C4 and C5: Aggregate shocks and composition effects

Read Table C4:

- Table C4 adds aggregate volatility to the model.
- The benchmark model already has no aggregate shock volatility, but the appendix asks whether aggregate shocks could substitute for network-generated volatility.
- Raising aggregate volatility can affect moments, but it does not replace the core role of networks in matching volatility dispersion.

Read Table C5:

- Table C5 studies whether changing public-firm composition can explain the empirical patterns.
- It compares data with a model of public firms with network data and a model using an alternative internal diversification function.

- The composition-only channel cannot replicate the large movements in volatility moments.

Economic message: The appendix exercises address two natural alternatives: common aggregate shocks and sample composition. Neither appears sufficient to replace the network mechanism.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Aggregate volatility (σ_{at})	...	.00	.01	.02	.03	.04	.05	.10	.20
Fundamental shock volatility (σ_e)	...	.298	.298	.298	.298	.298	.298	.298	.298
A. Firm Size and Volatility Distribution									
log size:									
Average	19.83	19.83	19.83	19.83	19.83	19.83	19.83	19.83	19.83
Standard deviation	2.56	2.53	2.53	2.53	2.53	2.53	2.53	2.53	2.53
50th-10th percentile	2.91	2.37	2.37	2.37	2.37	2.37	2.37	2.37	2.37
90th-50th percentile	3.88	3.91	3.91	3.91	3.91	3.91	3.91	3.91	3.91
log variance:									
Average	-1.40	-1.40	-1.33	-1.14	-.89	-.64	-.39	.62	1.86
Standard deviation	1.05	.86	.82	.74	.65	.56	.48	.25	.10
50th-10th percentile	1.38	.40	.36	.28	.21	.15	.11	.04	.01
90th-50th percentile	1.31	1.09	1.03	1.47	1.26	1.07	.89	.39	.12
B. Network Moments									
N^{**} :									
Median	1.00	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
log 50th-10th percentile	.40	.08	.08	.08	.08	.08	.08	.08	.08
log 90th-50th percentile	.83	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
H^{**} :									
Median	.05	.18	.18	.18	.18	.18	.18	.18	.18
log 50th-10th percentile	1.27	1.88	1.88	1.88	1.88	1.88	1.88	1.88	1.88
log 90th-50th percentile	1.85	1.71	1.71	1.71	1.71	1.71	1.71	1.71	1.71

Table C4, part 1: Aggregate-shock model moments.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
N^{**} :									
Median	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
log 50th-10th percentile	.60	.00	.00	.00	.00	.00	.00	.00	.00
log 90th-50th percentile	1.69	1.09	1.09	1.09	1.09	1.09	1.09	1.09	1.09
H^{**} :									
Median	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
log 50th-10th percentile	.95	.89	.89	.89	.89	.89	.89	.89	.89
log 90th-50th percentile	.90	.00	.00	.00	.00	.00	.00	.00	.00
C. Cross-Sectional Correlations									
Corr(log size, log var)	-.64	-.61	-.60	-.38	-.36	-.33	-.31	-.45	-.40
Corr(log N^{**} , log size)	-.00	.31	.31	.31	.31	.31	.31	.31	.31
Corr(log H^{**} , log size)	-.28	-.88	-.88	-.88	-.88	-.88	-.88	-.88	-.88
Corr(log H^{**} , log var)	.22	.51	.50	.49	.47	.45	.43	.38	.35
Corr(log N^{**} , log size)	.53	.34	.34	.34	.34	.34	.34	.34	.34
Corr(log H^{**} , log size)	-.47	-.34	-.34	-.34	-.34	-.34	-.34	-.34	-.34
Corr(log H^{**} , log var)	.13	.19	.19	.18	.17	.17	.16	.14	.12
D. Firms with Network Data									
log number of public firms	7.15	6.95	6.95	6.95	6.95	6.95	6.95	6.95	6.95
E. Goodness of Fit Statistics									
Function value	...	2.88	2.93	3.11	3.50	4.09	4.82	9.05	16.50
Wald	...	24.76	...	...	...	...	...	...	...
$\hat{p}$	...	.64	...	...	...	...	...	...	...

NOTE.—This table reports different size, variance, and network moments both from the data and from our model simulation. It takes the same structure as table 3. Column 1 reports the data. Column 2 repeats the estimation results for the benchmark model. The benchmark model has zero aggregate shock volatility ($\sigma_{at} = 0.00$). Columns 3-9 report simulations for a model with increasing aggregate volatility, keeping idiosyncratic shock volatility fixed at its benchmark value.

Table C4, part 2: Continued aggregate-shock model moments.

Year	DATA				MODEL			
	Public Firms with Network Data				Public Firms with Network Data			
	Number of Firms (1)	Size Dispersion (2)	Average Volatility (3)	Volatility Dispersion (4)	Number of Firms (5)	Size Dispersion (6)	Average Volatility (7)	Volatility Dispersion (8)
A. No Network Effects								
1981-87	.28	.21	.18	.24	.27	.21	.01	.02
1987-92	-.02	-.11	.27	.14	-.14	-.11	-.01	-.01
1992-2000	.16	.14	.59	-.24	.16	.14	.01	.02
2000-2004	-.19	-.28	-1.20	-.06	-.37	-.28	-.02	-.03
2004-7	.04	.02	1.34	-.17	.03	.02	.00	.00
2007-12	-.06	.01	-1.69	.13	.02	.01	.00	.00
B. Different Internal Diversification Function								
1981-87	.28	.21	.18	.24	.28	.21	.16	.04
1987-92	-.02	-.11	.27	.14	-.14	-.11	-.09	-.02
1992-2000	.16	.14	.59	-.24	.16	.14	.10	.03
2000-2004	-.19	-.28	-1.20	-.06	-.38	-.28	-.22	-.05
2004-7	.04	.02	1.34	-.17	.03	.02	.01	.00
2007-12	-.06	.01	-1.69	.13	.02	.01	.01	.00

NOTE.—Columns 1-4 report data moments, while cols. 5-8 report their model counterparts. All moments are percentage changes over the period listed. We report three moments from both the data and the model: size dispersion, average volatility, and volatility dispersion. We compute them using only public firms with network data available. Panel A is for our benchmark model. Panel B uses an alternative internal diversification function.

Table C5: Composition effect among public firms.

Parameter	Definition	Bench (1)	Alternative 1 (2)	Alternative 2 (3)
γ	Network propagation effect	.918 (.026)	.918	.913 (.011)
σ_e	Fundamental shock volatility	.298 (.035)	...	...
λ	Internal diversification effect	.131 (.025)	...	...
Z	New connections	.003 (.001)	.003	.003 (.000)
ψ	Sensitivity of connection weight to customer size	.184 (.003)	.184	.179 (.011)
μ_i	Mean of initial log size distribution	13.412 (.243)	13.412	13.456 (.144)

Table C6, part 1: Alternative internal-diversification parameters.

6.8 Tables C6 and C7: Alternative internal diversification

Read Tables C6 and C7:

- Table C6 reports parameters when internal diversification follows an alternative functional form inspired by Stanley et al. (1996).
- Table C7 reports the corresponding fit to size, variance, and network moments.
- Taking the Stanley et al. parameterization directly produces too much volatility and a poor fit.
- Re-estimating the alternative function improves the fit and leaves the network strength close to the benchmark.

Economic message: The specific functional form for internal diversification is not the central source of the paper's main conclusion. Even under an alternative specification, network effects remain important.

TABLE C6 (Continued)

Parameter	Definition	Bench (1)	Alternative 1 (2)	Alternative 2 (3)
σ_s	Standard deviation of initial log size distribution	5.104 (.102)	5.104 ...	5.089 (.086)
N_{pub}	Number of public firms	1,332 (47.7)	1,332 ...	1,321 (27.0)
κ		...	6.660 ...	2.249 (.300)
α		...	.150 ...	.097 (.005)

NOTE.—This table reports model parameters. Column 1 reports the estimated parameters from the SMM in the benchmark model. Column 2 holds the parameters constant at their benchmark values but uses an alternative internal diversification function due to Stanley et al. (1996). It takes the values for the parameters κ and α of that function directly from Stanley et al. (1996). Column 3 reestimates all parameters, including those of the internal diversification function.

Table C6, part 2: Continued
alternative-diversification parameters.

TABLE C7
SIZE, VARIANCE, AND NETWORK MOMENTS WITH ALTERNATIVE INTERNAL
DIVERSIFICATION FUNCTION

	DATA (1)	BENCHMARK MODEL (2)	ALTERNATIVE INTERNAL DIVERSIFICATION	
			No Reestimation (3)	Reestimation (4)
A. Firm Size and Volatility Distribution				
log size:				
Average	19.83	19.83	19.83	19.86
Standard deviation	2.56	2.53	2.53	2.52
50th–10th percentile	2.91	2.37	2.37	2.33
90th–50th percentile	3.88	3.91	3.91	3.90
log variance:				
Average	−1.40	−1.40	−.38	−1.31
Standard deviation	1.05	.86	1.18	1.01
50th–10th percentile	1.38	.40	.88	.93
90th–50th percentile	1.31	1.69	2.17	1.76
B. Network Moments				
N^{out} :				
Median	1.00	1.12	1.12	1.14
log 50th–10th percentile	.00	.08	.08	.10
log 90th–50th percentile	.83	1.02	1.02	1.00
H^{out} :				
Median	.05	.18	.18	.18
log 50th–10th percentile	1.27	1.88	1.88	1.87
log 90th–50th percentile	1.85	1.71	1.71	1.74
N^{in} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.00	.00	.00	.00
log 90th–50th percentile	1.69	1.09	1.09	1.09

Table C7, part 1: Alternative-diversification model
moments.

TABLE C7 (Continued)

	DATA (1)	BENCHMARK MODEL (2)	ALTERNATIVE INTERNAL DIVERSIFICATION	
			No Reestimation (3)	Reestimation (4)
H^{in} :				
Median	1.00	1.00	1.00	1.00
log 50th–10th percentile	.95	.89	.89	.88
log 90th–50th percentile	.00	.00	.00	.00
C. Cross-Sectional Correlations				
Corr(log size, log var)	–.64	–.61	–.69	–.78
Corr(log N^{out} , log size)	–.00	.31	.31	.31
Corr(log H^{out} , log size)	–.28	–.88	–.88	–.88
Corr(log H^{out} , log var)	.22	.51	.46	.61
Corr(log N^{in} , log size)	.53	.34	.34	.33
Corr(log H^{in} , log size)	–.47	–.34	–.34	–.33
Corr(log H^{in} , log var)	.13	.19	.22	.25
D. Firms with Network Data				
log number of public firms	7.15	6.95	6.95	6.94
E. Goodness of Fit Statistics				
Function value	...	2.88	3.76	2.29
Wald	...	24.76	...	25.13
p-value	...	.64	...	.62

NOTE.—This table reports different size, variance, and network moments both from the data and from our simulation. The structure is the same as table 3. Column 1 is for the data. Column 2 is for the benchmark model estimated by SMM. Columns 3 and 4 are for the alternative internal diversification function based on Stanley et al. (1996). Column 3 takes the values for the internal diversification function parameters κ and α of that function directly from Stanley et al. (1996). Column 4 reestimates all parameters, including those of the internal diversification function.

Table C7, part 2: Continued alternative-diversification model moments.

7 Short summary

- Firm volatilities comove strongly over time.

- The common factor in volatility is closely related to dispersion in the firm size distribution.
- In the market data, size dispersion is strongly correlated with both average firm variance and volatility dispersion.
- The same facts appear in sales-based fundamentals data.
- The model explains these facts through customer-supplier networks.
- Shocks propagate upstream from customers to suppliers.
- Large suppliers have more customers and therefore diversify better.
- Large customers receive larger weights in supplier customer bases.
- A supplier with a concentrated customer base has high out-Herfindahl and high volatility.
- The SMM estimation finds strong network propagation and shows that a no-network model cannot match volatility dispersion.
- Firm-level regressions show that size lowers volatility and customer concentration raises volatility.
- Factor regressions show that size dispersion explains a meaningful share of firm-volatility variation, especially for small firms.

8 Conclusion

8.1 Core conclusion

Herskovic, Kelly, Lustig, and Van Nieuwerburgh show that firm volatility is shaped by granular production networks. Firm size dispersion and network sparsity make firms' customer bases imperfectly diversified, so shocks to large customers can have outsized effects on suppliers. This generates both cross-sectional heterogeneity in volatility and common time-series movements in volatility.

8.2 Economic intuition

A firm is like a portfolio of its customers. If the firm has many similar-sized customers, customer shocks diversify away. If the firm has few customers or one dominant customer, the firm is exposed to large idiosyncratic shocks. When the entire economy becomes more granular, many firms' customer portfolios become less diversified at the same time, creating a common volatility factor.

8.3 Incremental contribution revisited

The paper's main contribution is to endogenize firm-level heteroscedasticity using networks. Instead of assuming that firms have different volatility for unspecified reasons, it links volatility to the size distribution and the network of customer-supplier relationships.

8.4 Implications

The findings imply that changes in market concentration can affect not only aggregate volatility but also the volatility of individual firms. They also imply that customer concentration is an economically meaningful risk characteristic, not just an accounting detail.

8.5 Limitations

The network data cover public firms and are truncated by customer-reporting rules. The estimated network parameter is large and requires careful interpretation. The model is intentionally parsimonious and abstracts from richer endogenous network formation, heterogeneous propagation mechanisms, and other sources of firm volatility.

8.6 Takeaway

The enduring takeaway is that firm volatility is a network outcome. Firm size dispersion creates a common volatility factor, while each firm's position in the customer-supplier network determines how strongly it is exposed to that factor.